

Úvod do R

Jana Kalická

2026-09-25

Adresar mozno nastavit aj v Session - Set Working Directory

```
getwd() # pracovny adresar
```

```
## [1] "E:/2026-2027/SADR/Cvicenia"
```

```
dir() #subory
```

```
## [1] "a.R"          "cv1.docx"      "cv1.html"
## [4] "cv1.R"        "cv2.docx"      "cv2.html"
## [7] "cv2.R"        "data_vyuka.xlsx" "diabetes.csv"
## [10] "export_studenti.csv" "export_studenti.xlsx" "cholesterol.xlsx"
## [13] "pomoc.R"      "Rplot.jpeg"    "sablon_1.html"
## [16] "sablon_pretty.html" "SADR1.html"      "SADR1.R"
## [19] "SADR1.spin.R"  "SADR1.spin.Rmd"  "SADR2.html"
## [22] "SADR2.R"
```

```
list.files()
```

```
## [1] "a.R"          "cv1.docx"      "cv1.html"
## [4] "cv1.R"        "cv2.docx"      "cv2.html"
## [7] "cv2.R"        "data_vyuka.xlsx" "diabetes.csv"
## [10] "export_studenti.csv" "export_studenti.xlsx" "cholesterol.xlsx"
## [13] "pomoc.R"      "Rplot.jpeg"    "sablon_1.html"
## [16] "sablon_pretty.html" "SADR1.html"      "SADR1.R"
## [19] "SADR1.spin.R"  "SADR1.spin.Rmd"  "SADR2.html"
## [22] "SADR2.R"
```

Matematicke operacie a funkcie

```
2+3
```

```
## [1] 5
```

```
2/3
```

```
## [1] 0.6666667
```

```
14 %% 3 #zvysok po deleni
```

```
## [1] 2
```

```
14 %/% 3 #celociselne delenie
```

```
## [1] 4
```

```

2^3

## [1] 8
sqrt(9)

## [1] 3
abs(-9)

## [1] 9
# Logaritmy
log(10)# prirodzeny logaritm

## [1] 2.302585
log10(10)# desiatkovy

## [1] 1
log(10,base=2)

## [1] 3.321928
log(0)# limita, ak ide zratat

## [1] -Inf
log(-1)# NaNs nie je definovana

## Warning in log(-1): NaNs produced
## [1] NaN
Exponencialna f.
exp(1)

## [1] 2.718282
vkladanie do premennej, klavesova skratka Alt -
a <- 5
a

## [1] 5
Faktorial a kombinacne cisla
factorial(5)

## [1] 120
choose(5,2)

## [1] 10

```

Datove typy objektov

typ double -dva podtypy, integer 5L, numeric 5

```

b <- 5L
c <- 5
class(b)

```

```
## [1] "integer"
```

```
class(c)
```

```
## [1] "numeric"
```

Zmena na iny typ

```
as.integer(c)
```

```
## [1] 5
```

Komplexne cisla

```
d <- 2+3i
```

```
class(d)
```

```
## [1] "complex"
```

Logicke hodnoty

```
5>7
```

```
## [1] FALSE
```

```
5<7
```

```
## [1] TRUE
```

```
5==7
```

```
## [1] FALSE
```

```
5!=7
```

```
## [1] TRUE
```

Datove struktury a vektory

```
v1 <- c(2,3,4)#vektor
```

```
class(v1)
```

```
## [1] "numeric"
```

```
v2 <- c(1,0,-1)
```

scitanie, delenie po zlozkach

```
v1+v2
```

```
## [1] 3 3 3
```

```
v1/v2
```

```
## [1] 2 Inf -4
```

aj pre vektory nerovnakej dlzky

```
v3 <- c(3,2)
```

```
v1+v3
```

```
## Warning in v1 + v3: longer object length is not a multiple of shorter object
## length
```

```
## [1] 5 5 7
```

funkčne hodnoty v každom bode vektora

```
log(v1)
```

```
## [1] 0.6931472 1.0986123 1.3862944
```

```
cos(v1)
```

```
## [1] -0.4161468 -0.9899925 -0.6536436
```

```
cos(v1*pi)
```

```
## [1] 1 -1 1
```

dlžka, súčet súčin, kumulovaný súčet

```
length(v1)
```

```
## [1] 3
```

```
sum(v1)
```

```
## [1] 9
```

```
cumsum(v1)
```

```
## [1] 2 5 9
```

```
prod(v1)
```

```
## [1] 24
```

Vektory ako postupnosti čísiel

```
1:10# postupnosť od prvého po posledný
```

```
## [1] 1 2 3 4 5 6 7 8 9 10
```

```
10:1
```

```
## [1] 10 9 8 7 6 5 4 3 2 1
```

```
seq(from=1, to=15,by=2)# s krokom 2
```

```
## [1] 1 3 5 7 9 11 13 15
```

v1 opakujem 4x

```
rep(v1,times=4)
```

```
## [1] 2 3 4 2 3 4 2 3 4 2 3 4
```

každý prvok 2x

```
rep(v1,each=2)
```

```
## [1] 2 2 3 3 4 4
```

voľba opakovania prvkov

```
rep(v1,times=c(2,3,4))
```

```
## [1] 2 2 3 3 3 4 4 4 4
```

```
rm(v3)
```

Matice

```
v3 <- 1:12
A <- matrix(v3,3,4)
A
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    4    7   10
## [2,]    2    5    8   11
## [3,]    3    6    9   12
```

```
B <- matrix(v3,3,4,byrow=T)
B
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    2    3    4
## [2,]    5    6    7    8
## [3,]    9   10   11   12
```

```
cbind(A,B) # spajanie po stlpkoch
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8]
## [1,]    1    4    7   10    1    2    3    4
## [2,]    2    5    8   11    5    6    7    8
## [3,]    3    6    9   12    9   10   11   12
```

```
rbind(A,B) #spajanie po riadkoch
```

```
##      [,1] [,2] [,3] [,4]
## [1,]    1    4    7   10
## [2,]    2    5    8   11
## [3,]    3    6    9   12
## [4,]    1    2    3    4
## [5,]    5    6    7    8
## [6,]    9   10   11   12
```

Datove tabulky a zoznamy, data.frame, tibble

Tabulka typu data.frame Nasa tabulka- student-meno,vek, stud. smer,body z pisomky

```
meno <- c("Jan","Iva","Eva","Oto","Ivo")
vek <- c(18,20,21,22,19)
studenti <- data.frame(meno,vek)
studenti
```

```
##  meno vek
## 1  Jan  18
## 2  Iva  20
## 3  Eva  21
## 4  Oto  22
## 5  Ivo  19
```

```
class(studenti)
```

```
## [1] "data.frame"

studenti$body <- c(10,12,15,0,20)
studenti

##   meno vek body
## 1  Jan  18   10
## 2  Iva  20   12
## 3  Eva  21   15
## 4  Oto  22    0
## 5  Ivo  19   20

studenti$studium <- factor(c(1,1,0,0,1),labels=c("iss","umint"))
studenti

##   meno vek body studium
## 1  Jan  18   10  umint
## 2  Iva  20   12  umint
## 3  Eva  21   15    iss
## 4  Oto  22    0    iss
## 5  Ivo  19   20  umint

studenti$pohlavie <- c("m","z","z","m","m")
str(studenti)

## 'data.frame':    5 obs. of  5 variables:
##  $ meno      : chr  "Jan" "Iva" "Eva" "Oto" ...
##  $ vek       : num  18 20 21 22 19
##  $ body      : num  10 12 15 0 20
##  $ studium   : Factor w/ 2 levels "iss","umint": 2 2 1 1 2
##  $ pohlavie : chr  "m" "z" "z" "m" ...

write.csv(studenti,"export_studenti.csv",row.names = F)
```

instalujeme writexl, kniznice, ktore este neboli pouzite sa vzdy najprv instaluju

```
library(writexl)
write_xlsx(studenti,"export_studenti.xlsx")
```

Filtrovanie cez prikaz subset

podmnoziny, len zeny, len umint

```
zeny <- subset(studenti,studenti$pohlavie=="z")
zeny

##   meno vek body studium pohlavie
## 2  Iva  20   12  umint          z
## 3  Eva  21   15    iss          z

umint <- subset(studenti,studenti$studium=="umint")
umint

##   meno vek body studium pohlavie
## 1  Jan  18   10  umint          m
## 2  Iva  20   12  umint          z
## 5  Ivo  19   20  umint          m
```

zeny a umint zeny alebo umint

```
z_umint <- subset(studenti, studenti$pohlavie=="z" & studenti$studium=="umint")
z_umint1 <- subset(studenti, studenti$pohlavie=="z" | studenti$studium=="umint")
z_umint1
```

```
##   meno vek body studium pohlavie
## 1 Jan  18  10  umint      m
## 2 Iva  20  12  umint      z
## 3 Eva  21  15   iss      z
## 5 Ivo  19  20  umint      m
```

Datova tabulka typu tibble

```
library(tibble)
st <- tibble(x=1:5, y=studenti$meno, z=studenti$vek)
st
```

```
## # A tibble: 5 x 3
##       x y      z
##   <int> <chr> <dbl>
## 1     1 Jan    18
## 2     2 Iva    20
## 3     3 Eva    21
## 4     4 Oto    22
## 5     5 Ivo    19
```

```
str(studenti)
```

```
## 'data.frame': 5 obs. of 5 variables:
## $ meno : chr "Jan" "Iva" "Eva" "Oto" ...
## $ vek : num 18 20 21 22 19
## $ body : num 10 12 15 0 20
## $ studium : Factor w/ 2 levels "iss","umint": 2 2 1 1 2
## $ pohlavie: chr "m" "z" "z" "m" ...
```

```
st1 <- as_tibble(studenti)
st1
```

```
## # A tibble: 5 x 5
##   meno   vek body studium pohlavie
##   <chr> <dbl> <dbl> <fct>   <chr>
## 1 Jan    18   10 umint    m
## 2 Iva    20   12 umint    z
## 3 Eva    21   15 iss      z
## 4 Oto    22    0 iss      m
## 5 Ivo    19   20 umint    m
```

Import dat

Data csv a xlsx

```
diabetes <- read.csv("diabetes.csv")
str(diabetes)
```

```
## 'data.frame': 768 obs. of 9 variables:
## $ Pregnancies : int 6 1 8 1 0 5 3 10 2 8 ...
```

```
## $ Glucose          : int  148 85 183 89 137 116 78 115 197 125 ...
## $ BloodPressure    : int   72 66 64 66 40 74 50 0 70 96 ...
## $ SkinThickness    : int   35 29 0 23 35 0 32 0 45 0 ...
## $ Insulin          : int    0 0 0 94 168 0 88 0 543 0 ...
## $ BMI              : num  33.6 26.6 23.3 28.1 43.1 25.6 31 35.3 30.5 0 ...
## $ DiabetesPedigreeFunction: num  0.627 0.351 0.672 0.167 2.288 ...
## $ Age              : int   50 31 32 21 33 30 26 29 53 54 ...
## $ Outcome          : int    1 0 1 0 1 0 1 0 1 1 ...
```

```
View(diabetes)
library(readxl)
data <- read_xlsx("data_vyuka.xlsx")
str(data)
```

```
## tibble [50 x 13] (S3: tbl_df/tbl/data.frame)
## $ respondent: num [1:50] 1 2 3 4 5 6 7 8 9 10 ...
## $ pohlavie  : chr [1:50] "m" "m" "m" "z" ...
## $ vek       : num [1:50] 28 22 50 39 53 55 50 39 57 45 ...
## $ vzdelanie : num [1:50] 3 2 1 2 3 1 3 1 2 2 ...
## $ stav      : num [1:50] 1 1 2 2 2 3 2 2 2 2 ...
## $ fajciar   : num [1:50] 0 0 0 0 0 0 0 1 0 0 ...
## $ praca     : num [1:50] 2 1 2 3 1 3 3 3 1 2 ...
## $ clen      : num [1:50] 5 4 4 5 5 2 3 4 2 5 ...
## $ mprij     : num [1:50] 1500 900 975 1200 1800 ...
## $ mstrava   : num [1:50] 250 200 350 400 400 200 350 250 300 320 ...
## $ rdovolenka: num [1:50] 10000 350 300 500 15000 200 900 200 250 450 ...
## $ pocit     : num [1:50] 73 27 65 26 38 19 42 57 10 62 ...
## $ invest    : num [1:50] 2 4 2 3 2 4 2 3 4 1 ...
```

```
View(data)
```

Statistická definícia pravdepodobnosti

10 x hadzem kockou, pozrieme ako padali hody, vieme, že pravdepodobnosť je 1/6 jeden konkrétny hod

```
hod <- sample(1:6,size=10, replace = T)
hod
```

```
## [1] 2 3 1 2 3 2 2 1 3 6
```

```
table(hod)
```

```
## hod
## 1 2 3 6
## 2 4 3 1
```

```
prop.table(table(hod))
```

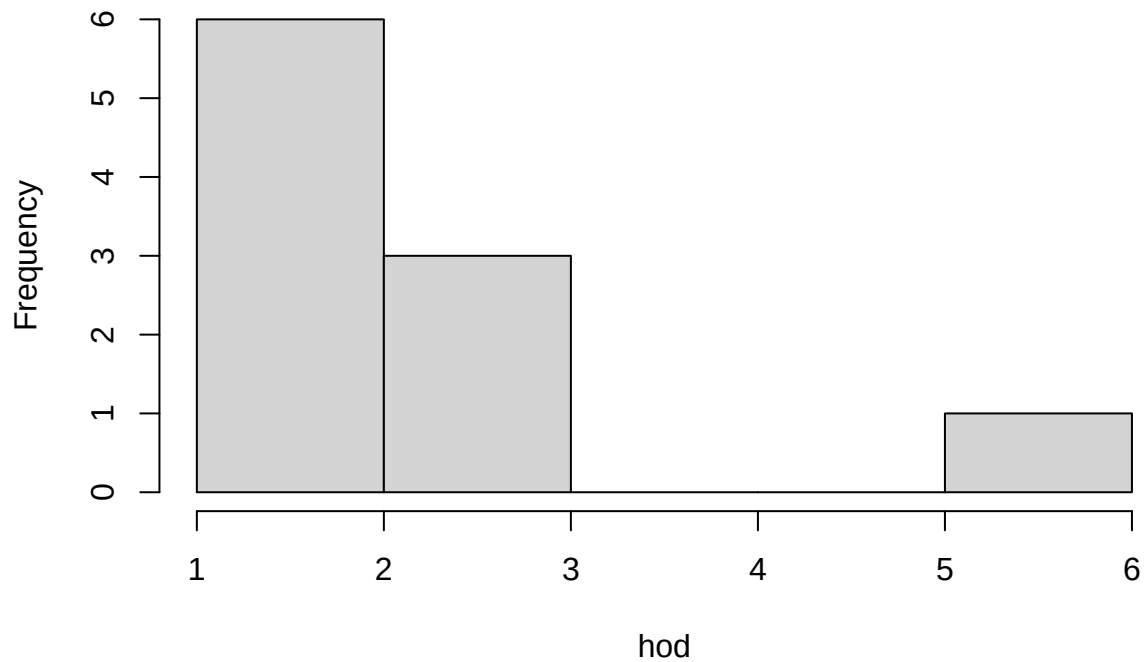
```
## hod
## 1 2 3 6
## 0.2 0.4 0.3 0.1
```

```
1/6
```

```
## [1] 0.1666667
```

```
hist(hod)
```

Histogram of hod

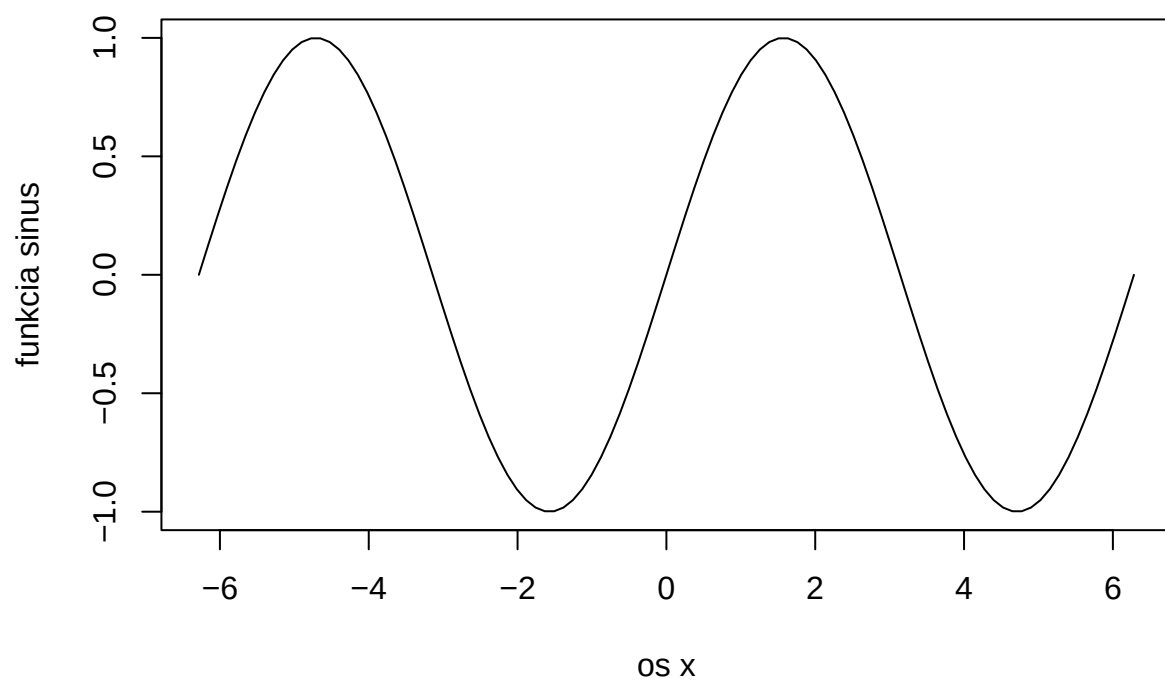


```
for (n in c(10, 100, 1000, 10000, 100000)) {
  print(prop.table(table(sample(1:6, size = n, replace = TRUE))))
}
```

```
##
##  1  2  3  4  5  6
## 0.1 0.2 0.2 0.2 0.2 0.1
##
##  1  2  3  4  5  6
## 0.14 0.13 0.16 0.17 0.23 0.17
##
##  1  2  3  4  5  6
## 0.161 0.176 0.155 0.159 0.168 0.181
##
##  1  2  3  4  5  6
## 0.1686 0.1646 0.1671 0.1726 0.1634 0.1637
##
##  1  2  3  4  5  6
## 0.16434 0.16571 0.16807 0.16835 0.16790 0.16563
```

Jednoduche grafy

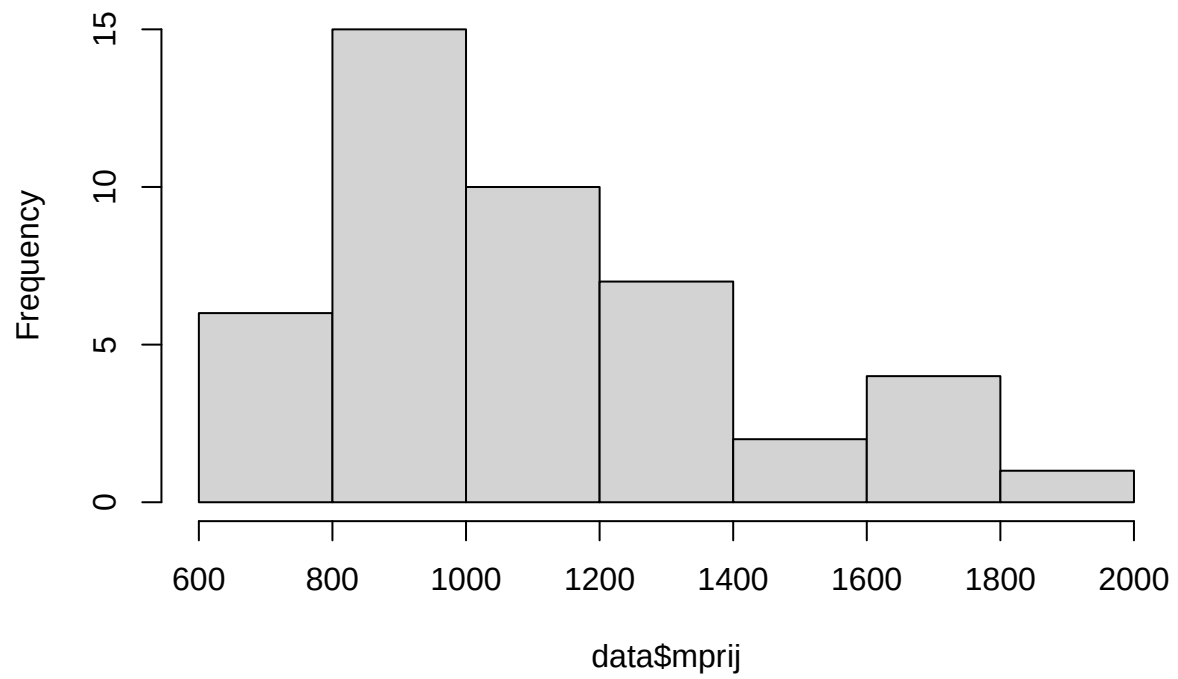
```
plot(sin, from= -2*pi, to= 2*pi, xlab="os x", ylab="funkcia sinus")
```



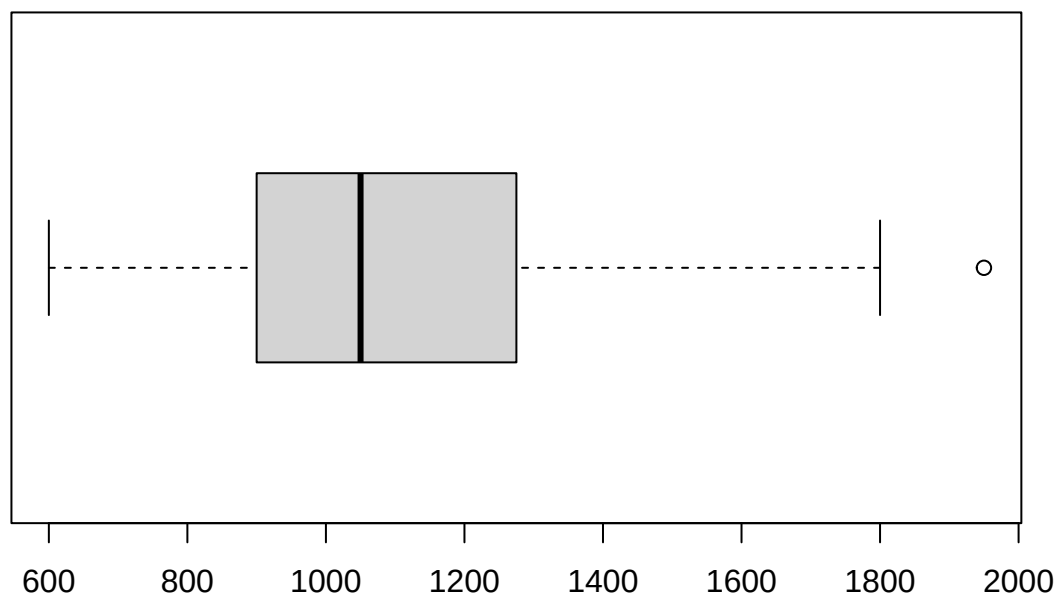
Histogram a krabicovy graf

```
hist(data$mprij)
```

Histogram of data\$mprij



```
boxplot(data$mprij, horizontal = T)
```



```
boxplot(data$mprij~data$pohlavie, horizontal = T)
```

